4 A rod is attached vertically to a horizontal turntable at point **P**, 0.080 m from the centre **Q**, as shown in Fig. 4.1.

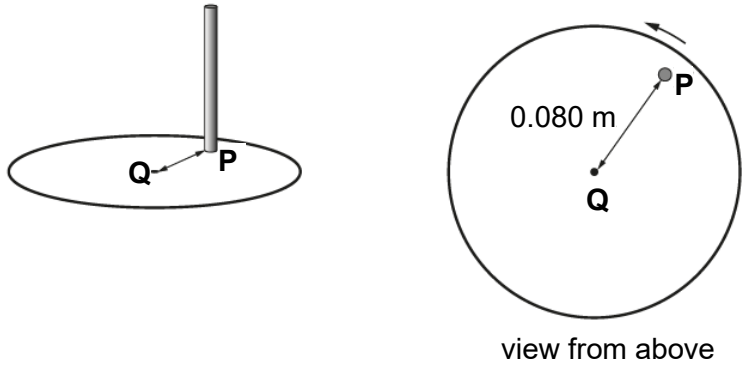

Fig. 4.1

(a) The turntable rotates around **Q** at 45 rotations per minute. Calculate

(i) the angular velocity of the rod,

angular velocity of rod = ...rad s^{-1} [2]

(ii) the acceleration of the rod and state its direction.

acceleration of rod = ...m s^{-2} [2]

direction of acceleration= ..... [1]

(b) When the rod is illuminated from the side, its shadow on a screen oscillates.

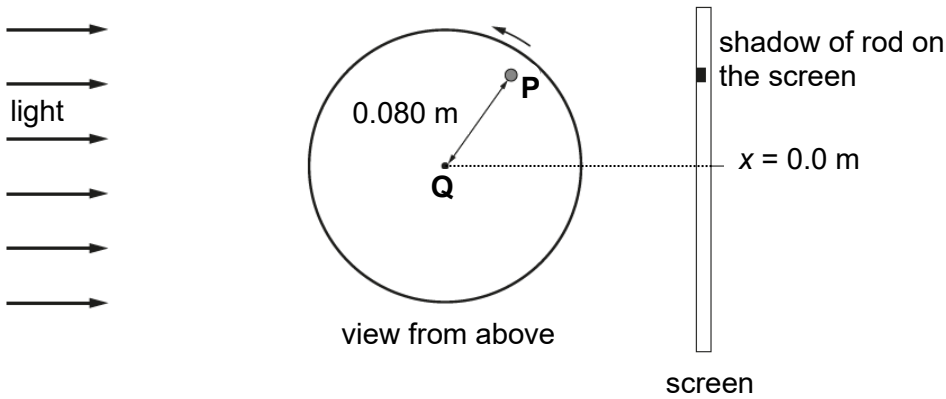

Fig. 4.2

The displacement x in metres of the centre of the shadow from the centre of oscillation and the acceleration a of the shadow may be written

$$x = A\sin(\omega t)$$

where t is time in seconds, A is the amplitude of oscillation and ω is the angular velocity of the rod.

(i) Using the values given in **(a)**,

1. calculate the period of oscillation of the shadow, and

period of oscillation = ... s [1]

2. derive an expression for the acceleration of the shadow in terms of t.

expression for acceleration = ... [2]

(ii) Determine the acceleration of the shadow at t = 0.20 s.

acceleration =.....m s^{-2} [2]

(iii) In Fig. 4.3, sketch the variation of acceleration of the oscillation with time for at least two cycles. Label the necessary values.

[3]

Fig. 4.3

(iv) Fig. 4.4 shows a second rod, which is now attached vertically to the turntable at point **R**, 0.060 m from centre **Q**, such that angle **PQR** is a right angle.

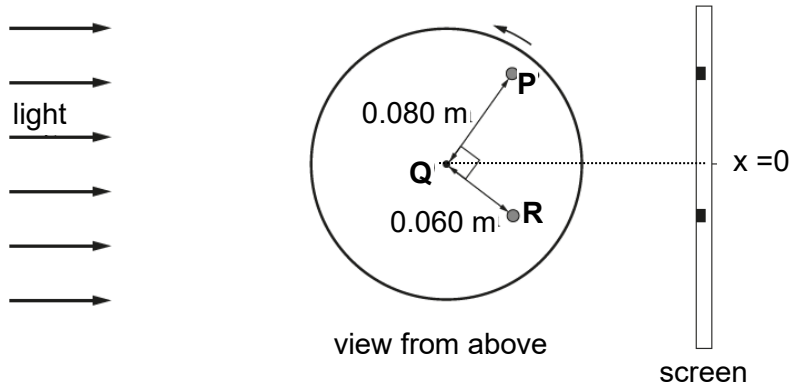

Fig. 4.4

If the expression for displacement of the shadow of the first rod at **P** in **(b)** is $A\sin(\omega t)$, derive an expression for displacement of the shadow of the second rod at **R** with the necessary values of its amplitude and ω.

expression for displacement = ... [2]